# Jing Xiang Xu

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## EDUCATION

**Cornell University**, College of Engineering, Ithaca, NY
Master of Engineering in Financial Engineering **Expected December 2027**

**The University of Hong Kong**, Hong Kong
Bachelor of Science in Mathematics with a Minor in Computer Science, **First Class Honors**, **GPA: 3.74** **June 2026**

**University of Oxford**, Oxford, United Kingdom
Visiting Student Program in Mathematics (ROA scholarship) **July 2025**
*Selected Coursework*: Data Structures and Algorithms, Convex Optimization, Machine Learning

## SKILLS

Technical: Python, C++, R, SQL, Shell scripting, Git, Linux
Quantitative: Index Rebalancing, Backtesting
AI/ML: Prompt engineering, DeepSeek-based classification and reranking, fine-tuning workflows, loop engineering

## EXPERIENCE

**Quantitative Researcher**, *Dymon Asia Capital (HK) Limited*, Hong Kong **May to Aug. 2026**

- Researched Hang Seng Composite and Tech Index rebalancing rules to predict constituent inclusions and exclusions before official announcements — a repeatable, thesis-driven edge built from public filings and rulebook analysis.
- Designed an auditable, fully reproducible Python research pipeline integrating FactSet, ERD and HKEX filings, including LLM-assisted screening gates and a filing-driven free-float engine, so every signal input can be traced to its source.
- Achieved an 82.6% action-level hit rate (185 of 224 additions and deletions) across the latest three HSCI rebalances in leakage-controlled, point-in-time evaluation, with no licensed index feed.

**Quantitative Researcher Intern**, *Rivermap Company Limited*, Hong Kong **Feb. to Apr. 2026**

- Automated thematic stock selection by mapping unstructured investment themes onto supply-chain dimensions and a proprietary sector taxonomy, turning qualitative theses into testable, auditable universes.
- Built a Python/Pandas pipeline integrating sector, revenue-segment and company-description data, generating auditable intermediate outputs and segment-level relevance diagnostics for every screen.
- Designed a two-stage semantic retrieval pipeline (BGE embeddings with cosine similarity, then LLM-based refinement) for candidate generation and ambiguous theme resolution.

**Quantitative Strategy Analyst Intern**, *Nine Martingale Investment Management, Ltd.*, Shanghai **July to Sep. 2025**

- Identified a 60-day dividend-event data lag and corrected factor alignment, raising information coefficient by 0.02.
- Built a factor-backtesting framework; improved annualized return by 6% and Sharpe ratio by 0.3 over five years.
- Analyzed daily exposures to 10 Barra style factors to characterize signal crowding, decay and style tilts.

**Quantitative Strategy Analyst Intern**, *Northeast Securities Co., Ltd.*, Shanghai **July to Sep. 2024**

- Developed mean-variance and risk-parity models within a 15-year daily-rebalanced backtesting framework.
- Implemented a Monte Carlo method approximating ERC weights by minimizing asset risk-contribution gaps.

## PROJECTS

**Researcher,** *HKU Quant Trading Team*, Hong Kong **Feb. to Aug. 2026**

- Analyzed historical Polymarket data for systematic price-versus-outcome gaps, grouping contracts by topic, market structure and time to resolution, and flagging Yes/No shares with pricing gaps above 3 percentage points.
- Built a Python pipeline that scanned live markets, ranked opportunities by expected value, enforced position limits, submitted orders through the order book, and tracked realized P&L.

**Independent Researcher**, *University of Oxford*, Oxford, United Kingdom **Jan. to June 2025**

- Modeled Lightning Network channel management as a discounted MDP, deriving optimal routing, liquidity-reset and channel-capacity policies under stochastic payment flows.
- Implemented value iteration for discrete and exponential payment distributions, proving bidirectional flows can naturally rebalance liquidity and remove costly resets.

## ACTIVITIES/INTERESTS

HKU Young Scientist Scheme; F1 racing; Texas Hold'em poker; running